

Meridian Compiler Toolchain - Release Notes 4.2

Build reproducibility

Identical inputs now produce byte-identical artifacts. The build ID is derived from a hash of the source tree rather than from the wall clock, so two machines building the same commit agree bit for bit.

Incremental linking

Link times on the reference workload dropped from 41 seconds to 6 seconds by caching relocation tables between builds. The cache is keyed on object file content hashes and lives under `.meridian/linkcache`.

Diagnostics

Error messages now include the inferred type of every subexpression in a failing template instantiation, instead of only the outermost type.

Known limitations

The new linker cannot yet emit split debug information on 32-bit targets. Use the legacy linker with the `--split-debug` flag there.